Clifford-Algebra-Based Geometric Feature Fusion for Real-Time UHD Low-Light Image Enhancement

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Abstract

Considering efficiency, ultra-high-definition (UHD) low-light image restoration is extremely challenging. Existing methods based on Transformer architectures or high-dimensional complex convolutional neural networks often suffer from the "memory wall" bottleneck, failing to achieve millisecond-level inference on edge devices. To address this issue, we propose a novel real-time UHD low-light enhancement network that fuses geometric features using Clifford algebra in 2D Euclidean space. First, we construct a four-layer feature pyramid with gradually increasing resolution, which decomposes input images into low-frequency and high-frequency structural components via a Gaussian blur kernel, and adopts a lightweight U-Net based on depthwise separable convolution for dual-branch feature extraction. Second, to resolve structural information loss and artifacts from traditional high-low frequency feature fusion, we introduce spatially aware Clifford algebra, which maps feature tensors to a multivector space (scalars, vectors, bivectors) and uses Clifford similarity to aggregate features while suppressing noise and preserving textures. In the reconstruction stage, the network outputs adaptive Gamma and Gain maps that perform physically constrained non-linear brightness adjustment according to Retinex theory. Integrated with FP16 mixed-precision computation and dynamic operator fusion, our method achieves millisecond-level inference for 4K/8K images on a single consumer-grade device, while outperforming state-of-the-art (SOTA) models on several restoration metrics.

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CCS Concepts

• **Computing methodologies** → **Computer vision problems**.

Keywords

UHD low- light image, memory wall, Clifford algebra, Gamma and Gain maps, Retinex theory.

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1 Introduction

While Ultra-High-Definition (UHD) imaging (e.g., 4K and 8K) is becoming standard in fields like autonomous driving and surveillance and modern computer vision tasks demand preservation of microscopic details [22], capturing these images in low-light environments remains highly challenging. Due to limited exposure times, conventional sensors often produce UHD images severely degraded by brightness attenuation, color distortion, and dense burst noise [7, 49].Most existing low-light enhancement algorithms are heavily optimized for standard-definition inputs (e.g., 1080p) [44]. Extending these tasks to extreme 4K and 8K resolutions is not a simple engineering adjustment, but a rigorous test of architectural design: Networks must achieve high-fidelity physical restoration without exceeding the stringent memory and computational limits imposed by massive pixel arrays[29, 58].

When applied to native UHD inputs, current state-of-the-art models generally face one of two major bottlenecks:

Heavyweight networks hit a memory wall: While Transformer architectures (e.g., LLFormer [41], HVI-CIDNet [49]) and

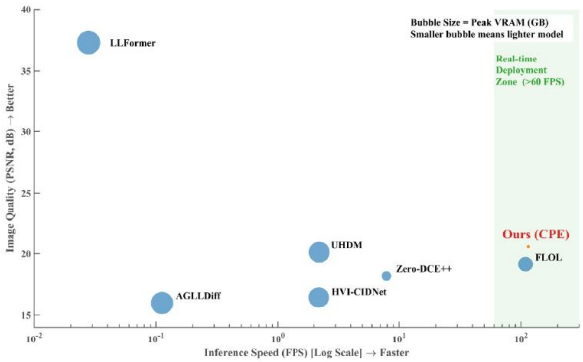


Figure 1: Comprehensive performance comparison of low-light enhancement models at 4K resolution. The x-axis and bubble size indicate inference latency and peak VRAM consumption (in log scale), respectively. The y-axis represents the reconstruction quality (PSNR). The green shaded area highlights the real-time processing regime (>30 FPS).

diffusion models (e.g., AGLLDiff [28]) excel at lower resolutions, their quadratic computational complexity makes direct UHD processing prohibitive. To bypass Out-of-Memory (OOM) errors, these models often rely on patch-based inference. This workaround, however, breaks global illumination consistency and frequently produces boundary artifacts [25], whereas forced global downsampling irreversibly destroys high-frequency textures.

Lightweight curve estimation lacks noise robustness: Conversely, parameterized curve-mapping networks like Zero-DCE++ [26] offer extreme inference speeds. Yet, because they lack explicit frequency decoupling and spatial denoising mechanisms, they tend to amplify dark-region noise alongside the signal. In extreme low-light scenarios, this inherently results in severe color washout and visual artifacts.

In response to these architectural bottlenecks, we propose Clifford Pyramid Enhance (CPE), an ultra-fast low-light enhancement framework designed for native UHD inputs (Figure 1). Rather than directly predicting massive RGB pixel arrays, CPE constructs a lightweight multi-scale pyramid driven by a "low-resolution feature extraction, high-resolution parameter mapping" strategy [19, 26, 53].

Since high-frequency details are essentially vector fields with directionality, while low-frequency illumination is a smooth scalar field, traditional linear concatenation will forcibly mix and destroy their geometric topology. To address this, at the heart of our framework is a novel feature fusion mechanism based on 2D Euclidean Clifford Algebra, Cl(2,0) [4, 23, 36]. While CPE initially separates low-frequency illumination from high-frequency details to isolate distinct degradations [20, 25], conventional fusion of these branches (via addition or concatenation) inevitably leads to structural degradation. We overcome this by projecting the decoupled feature tensors into a Clifford multivector space comprising scalars, vectors, and bivectors. This geometric transformation empowers the network to measure the spatial directionality of color and texture manifolds. By enforcing a geometric similarity constraint, CPE achieves a physically consistent feature aggregation that sharply retains local textures while robustly suppressing dark-region noise.

Finally, to map these latent features back to the UHD space without triggering OOM errors, CPE replaces traditional residual prediction with a Retinex-inspired enhancement mechanism [33, 50]. The model estimates spatially adaptive Gamma and Gain parameters at the raw resolution [2, 26], selectively applying non-linear enhancement solely to the illumination component. By proportionally scaling the original image using the ratio of enhanced to initial illumination, CPE restores exposure while inherently securing color fidelity. Because the high-frequency reflectance component remains intact, this approach fundamentally sidesteps the boundary artifacts common in patch-based processing [41].

In summary, the main **contributions** of our paper are three-fold:

i) To overcome the "memory wall" bottleneck typical of large models at 4K/8K resolutions, we construct a lightweight pyramid paired with a dynamically adaptive enhancement system. By performing feature extraction in a fixed, downsampled latent space (e.g., 1/4 scale) and executing parameterized non-linear mapping at the native resolution end based on Retinex theory, our architecture effectively prevents patch artifacts and detail degradation.

ii) We firstly introduce a 2D Clifford multivector space as a computationally efficient alternative to self-attention and simple channel concatenation in low-light enhancement. By using spatial directionality to gauge the geometric correlation between high- and low-frequency features, this module effectively suppresses dark-region noise while faithfully reconstructing the intricate high-frequency textures and pure colors inherent in UHD images.

iii) Extensive experiments on authoritative benchmarks (e.g., UHD_4K and UHD_8K) demonstrate that our method achieves state-of-the-art results for UHD enhancement in key metrics such as SSIM and color fidelity, alongside real-time inference speed. Clocking an average latency of 8.68 ms (115.2 FPS) at 4K and 11.49 ms (87.0 FPS) at 8K, our approach offers a highly practical solution for real-time UHD enhancement on resource-constrained edge devices.

2 Related Work

UHD Low-Light Image Enhancement Low-light image enhancement has evolved from traditional Retinex priors to sophisticated deep learning. Early mainstream efforts focused on end-to-end mapping [52, 53] or explicit Retinex decomposition [38, 40, 44, 57]. While later efficient paradigms like Zero-DCE++ [26] improved processing speed via curve estimation, these models often falter under extreme darkness—struggling with noise amplification and color shifts. This limitation has prompted a necessary transition toward physics-informed noise modeling [16, 24].

As 4K and 8K media become the norm, enhancing Ultra-High-Definition (UHD) content is now a critical frontier. While Transformers [41, 49], diffusion models [28], and recent wavelet-based methods such as Wave-Mamba [61] deliver stunning results, they often hit a "computational wall"; their complexity scales too aggressively for real-time 8K processing. To resolve this efficiency bottleneck, we propose a lightweight U-Net backbone using depthwise separable convolutions coupled with dynamic resolution recovery, effectively balancing visual quality and inference speed.

Multi-Scale Feature Fusion and Geometric Representation Efficient feature fusion is the backbone of high-quality image reconstruction. Traditional paradigms like FPN [27] and U-Net [35]

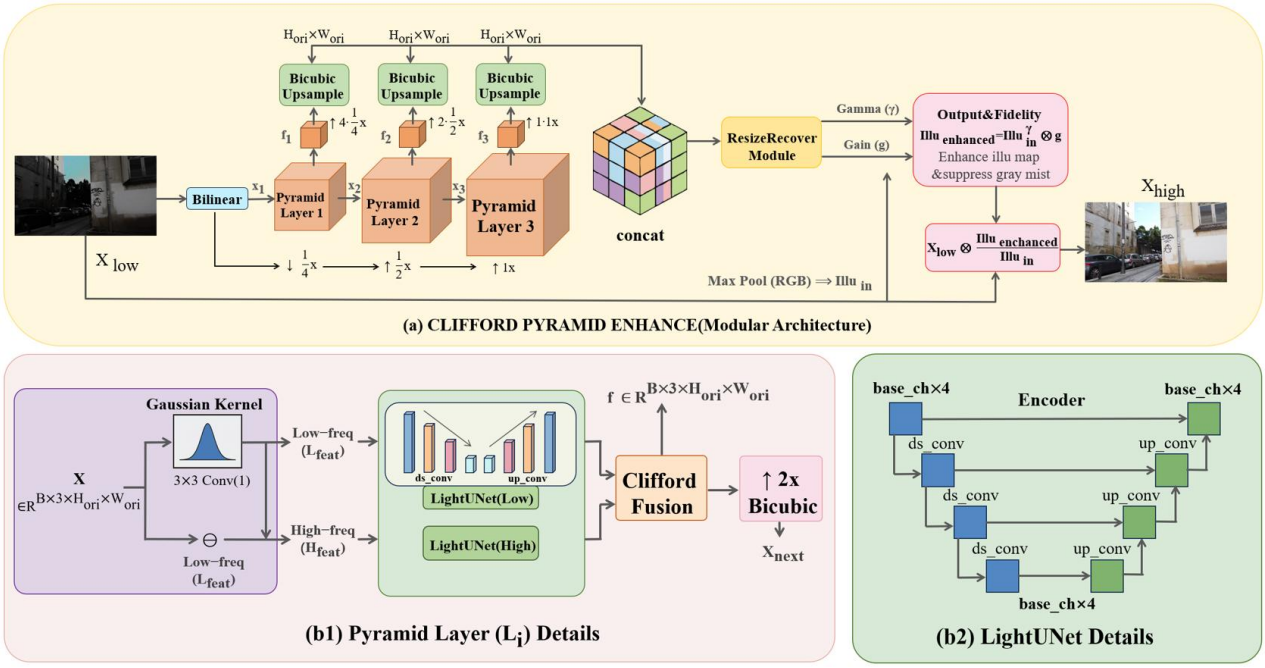


Figure 2: Overview of the proposed Clifford Pyramid Enhance (CPE) framework. (a) The overall pipeline based on a multi-scale pyramid. (b1) The frequency-decoupled feature extraction module (Pyramid Layer) for separating low-frequency illumination and high-frequency structural features. (b2) Details of the dual-branch LightUNet.

typically rely on simple concatenation or addition—linear operations that, while straightforward, often ignore the rich geometric correlations hidden between feature channels. To address this, Quaternion Neural Networks (QNNs) [60] and their recent variants [59] treat RGB channels as inseparable quaternions. By doing so, they capture internal color dependencies far more effectively than standard convolutions, significantly curbing color shifts.

However, quaternions primarily "see" dependencies within the color dimension, leaving the spatial gradients and directional geometry of image features largely untapped. To unlock higher-order geometric logic, researchers have begun integrating Clifford Algebra into deep learning [23, 36]. While Clifford convolutions have shown promise in processing complex 3D data and vector fields [4, 5], their potential for UHD low-light enhancement—specifically for fusing high-frequency structures with low-frequency illumination—remains uncharted territory. Building upon these advances, we introduce the 2D Euclidean Clifford Algebra, Cl(2,0), into this domain. Bypassing the heavy overhead of self-attention mechanisms, we formulate a frequency-adaptive fusion approach that explicitly models geometric correlations between high-frequency structures and low-frequency illumination.

3 Methodology

3.1 Overall Architecture

Figure 2 (a) shows our Clifford Pyramid Enhance (CPE) architecture [4]. A native UHD low-light image $X_{low} \in \mathbb{R}^{H_{ori} \times W_{ori} \times 3}$ is often too large to process directly—out-of-memory is an issue. To mitigate this, CPE first adaptively downsamples the input to a fixed spatial size (e.g.,64×64), yielding a compact base tensor X_{base}. From there, we build an inverse feature pyramid [54]. The network progressively upsamples X_{base} through three scales (e.g., $64 \times 64 \rightarrow 128 \times 128 \rightarrow 256 \times 256$). At each level i, a dedicated Pyramid Layer first decouples the intermediate signal to separate global illumination patterns from local high-frequency details. Instead of keeping them separate, the layer immediately fuses these components back together using a geometry-aware mechanism, outputting a unified, scale-specific feature map denoted as f_i. To restore the original resolution for lossless enhancement, we bicubically upsample $\{f_i\}_{i=1}^{3}$ to $H_{ori} \times W_{ori}$ and concatenate them along the channel dimension. Instead of predicting an enhanced image directly, this rich feature block is fed into the Dynamic Resolution Enhancement Module (Section 3.4) to generate adaptive parameters. By keeping the heavy computation in low-resolution space and only performing the final parameter mapping at full resolution [19, 26], the boundary artifacts that typically occur in patch-based methods are avoided [41, 46].

3.2 Frequency-Decoupled Feature Extraction

At each pyramid level, given that image degradation in extremely dark or adverse weather scenarios exhibits pronounced frequency differences [20, 25, 30]—global illumination and color shifts in the low-frequency band, severe noise and edge details in the high-frequency band—we design a Frequency-Decoupled Feature Extraction module. As mentioned in Section 3.1, this is the dedicated layer that explicitly separates these two signals (Figure 2 (b1)).

Specifically, given an input tensor X_{in}, the module applies a 3×3 Gaussian kernel with a fixed standard deviation to extract the low-frequency component L_{feat}. A residual subtraction ($H_{feat} = X_{in} - L_{feat}$) [25] then isolates the high-frequency component H_{feat}

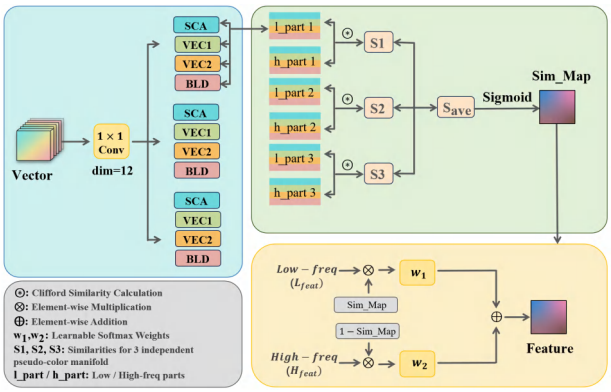


Figure 3: Details of the geometric feature fusion module based on Clifford Algebra Cl(2,0).

[42]. The decoupled components then go through two parallel lightweight U-Net (LightUNet) branches, which process them separately. For computational efficiency, LightUNet utilizes Depthwise Separable Convolutions (Figure 2 (b2)) [9, 52], expanding the receptive field to capture broader contextual information while cutting the parameter scale by an order of magnitude. Finally, the outputs from both branches are passed directly to the Clifford Fusion module (Section 3.3), which geometrically fuses them into a unified, scale-specific feature map f_i.

3.3 Clifford Geometric Feature Fusion

Multivector Space Construction: Figure 3 illustrates our Clifford Geometric Feature Fusion module. High-frequency components are crucial for structural details, while low-frequency components capture global context [1, 8, 13]. To preserve this inherent spatial directionality during fusion and circumvent structural artifacts caused by conventional channel concatenation [9, 53], we introduce 2D Euclidean Clifford Algebra, $Cl(2,0)$, to construct a geometry-aware aggregation module [4, 23, 36].

Specifically, instead of simple channel concatenation, we first project the deeply extracted low-frequency global features L_{feat} and high-frequency detail features H_{feat} into a 12-channel tensor space via 1×1 convolutions. This space is then reshaped into three independent pseudo-color manifolds (each with 4 channels), which form our Clifford algebra space. Within the $Cl(2,0)$ algebraic framework following standard Clifford algebra formulations [4], the 4-channel feature at each spatial location for a given manifold is rigorously mapped to a multivector M:

$$M = s + v_1 e_1 + v_2 e_2 + b e_{12}, \tag{1}$$

where s denotes the scalar component (characterizing base energy), v_1 and v_2 are orthogonal vector components (capturing the directionality of spatial gradients), and b represents the bivector (encoding rotational and areal attributes [36]).

Geometric Similarity Measurement and Dynamic Aggregation: Operating within this multivector space, we employ the Clifford inner product to compute the spatial correlation between high- and low-frequency features across the geometric manifold [5] via the geometric product, the fundamental operation of Clifford algebra. Specifically, we extract the scalar part, denoted by $\langle\cdot\rangle_0$, from the geometric product of M_L and the reversion of M_H (denoted as $M_H^\dagger$). To obtain a globally robust structural similarity mask, we average the raw spatial correlation across all three pseudo-color manifolds and apply a Sigmoid normalization function σ:

$$S_{map}(x,y) = \sigma\left(\frac{1}{3}\sum_{i=1}^{3}\langle M_{L,i}(x,y)M_{H,i}^{\dagger}(x,y)\rangle_0\right), \tag{2}$$

This yields a spatially adaptive similarity map $S_{\text{map}} \in [0,1]^{H\times W}$. which the module then uses to execute a dynamic aggregation operation:

$$F_{\text{fusion}} = w_1 \cdot L_{\text{feat}} \cdot S_{\text{map}} + w_2 \cdot H_{\text{feat}} \cdot (1 - S_{\text{map}}), \tag{3}$$

where w_1 and w_2 are learnable scalar weights strictly normalized via a Softmax function. Through this mechanism, the network effectively preserves high-frequency structural information in regions with consistent directionality (e.g., sharp edges). In contrast, in areas with chaotic directionality (e.g., snow occlusion or dark-region noise), it utilizes low-frequency components to suppress degradation, consistent with recent findings on directional consistency for robust adverse-weather restoration [10]. This approach significantly enhances the robustness of feature fusion while elegantly sidestepping the misalignment issues between high- and low-frequency information.

3.4 Dynamic Resolution Reconstruction and Color Fidelity

Resolution-Agnostic Parametric Mapping: After acquiring the fused multi-scale features, the network enters the final image enhancement stage. To decouple computational overhead from the UHD input resolution [17, 51], we build a Dynamic Resolution Reconstruction module. As illustrated in Figure 2 (a), the multi-scale features are upsampled and concatenated before being fed into this module. Following the efficient paradigm of curve-based and illumination-map estimation [26, 33, 45], the network outputs two sets of spatially adaptive parameter maps that match the native input resolution: Gamma features F_γ and Gain features F_{gain}.

To ensure proportional scaling across the RGB channels, we compute their mean along the channel dimension. Then, using a Sigmoid function, we generate the Gamma map Γ and Gain map G :

$$\Gamma = 0.15 + 0.85 \cdot \sigma(\text{mean}(F_\gamma)), \tag{4}$$

$$G = 0.8 + 1.7 \cdot \sigma(\text{mean}(F_{\text{gain}})), \tag{5}$$

where σ denotes the Sigmoid function. Rather than absolute physical constants, these bias terms and scaling factors are empirical heuristics guided by Retinex theory [6, 33]. For Gamma, we set $\Gamma \in (0.15, 1.0)$: the 0.15 lower bound acts as an engineering trade-off that enables effective non-linear contrast stretching for extreme dark regions while strictly avoiding color posterization and division-by-zero risks, whereas the 1.0 upper bound seamlessly preserves well-exposed structures. For Gain, we set $G \in (0.8, 2.5)$: the 0.8 minimum endows the model with the flexibility to slightly attenuate local overexposure (e.g., nocturnal halos), while the 2.5 maximum serves as a stable prior to compensate for severe global illumination attenuation without triggering gradient explosion.

Retinex-Based Color-Fidelity Reconstruction: Subsequently, we perform full-resolution, color-faithful rendering. First, guided by standard illumination priors [15, 18], the maximum channel value of

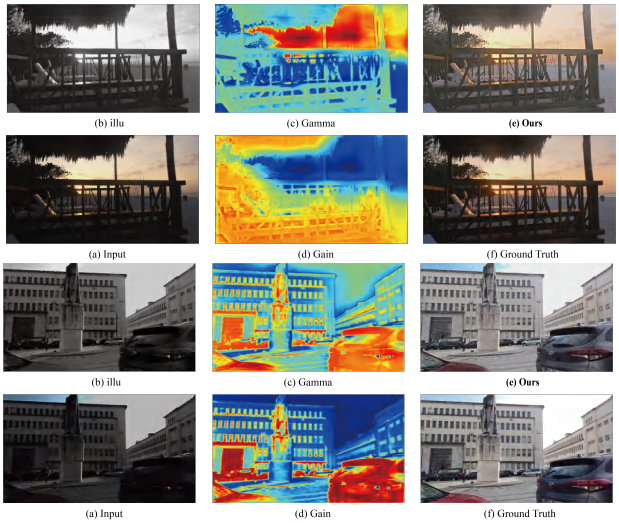


Figure 4: Dynamic resolution reconstruction and adaptive parameter mapping based on Retinex theory. (a) Original input. (b) Extracted initial illumination map. (c) Predicted Gamma map. (d) Predicted Gain map. (e) Enhanced output. (f) Ground Truth.

the original input image X_{low} is extracted as the initial illumination component $Illu_{\text{in}} = \max_{c\in\{R,G,B\}} X_{\text{low}}^{(c)}$. This initial illumination is then non-linearly adjusted using the predicted Γ and G:

$$Illu_{\text{enhanced}} = G \cdot Illu_{\text{in}}^{\Gamma}. \tag{6}$$

Here, the adaptive gamma Γ dynamically suppresses haze and enhances contrast based on the degradation degree of different regions, while the adaptive gain G handles global brightness elevation. Finally, to address the color shift issue that frequently occurs in traditional deep learning methods, we maintain the constant reflectance assumption [6, 38] and perform color-fidelity restoration on the ratio of the enhanced illumination to the initial illumination:

$$X_{\text{enh}} = X_{\text{low}} \odot \frac{Illu_{\text{enhanced}}}{Illu_{\text{in}}}. \tag{7}$$

This mechanism mathematically guarantees that the proportional relationship among the RGB channels remains strictly constant (i.e., channels balanced and fidelity preserved). Thanks to these simple multiply-add operations, our model achieves exceptionally high inference efficiency (87.0 FPS at 8K, 115.2 FPS at 4K) at full-frame UHD resolutions while maintaining exemplary color fidelity.

4 Experiments and Analysis

4.1 Experimental Setup

Datasets: We extensively evaluate our method on the authoritative UHD-LOL dataset [41], which comprises real-world low-light and normal-light image pairs. Following standard protocols, the 4K subset utilizes 5,999 pairs for training and 2,100 for testing, while the 8K subset uses 2,029 for training and 937 for cross-scale stress testing.

Baselines: We compare CPE against 7 state-of-the-art methods spanning four paradigms: heavy Transformers (LLFormer [41], Retinexformer [6]), scale-aware CNNs (FLOL [3], UHDM [51]), curve estimation (HVI-CIDNet [49], Zero-DCE++ [26]), and diffusion models (AGLLDiff [28]). For models suffering from Out-of-Memory (OOM) errors during native 4K/8K testing, we adopt their officially recommended mitigation strategies (e.g., overlapping patching or spatial downsampling) to capture their upper bounds under deployable conditions.

Implementation details: Our model is implemented in PyTorch and trained on a single NVIDIA RTX 3090 GPU (24GB VRAM). Training runs for 200 epochs using the AdamW optimizer [32] (initial lr 1×10^{-3} decaying to 1×10^{-6} via Cosine Annealing [31], weight decay 1×10^{-4}). Input pairs are randomly cropped to 512×512 patches with a batch size of 4. Data augmentation includes random geometric flips, dynamic hue perturbation ($[-0.1, 0.1]$), and saturation shifts ($[0.8, 1.2]$). The training objective is $L_{\text{total}} = L_{\text{Charbonnier}} + 0.5L_{\text{SSIM}} + 0.5L_{\text{Perceptual}}$, where L_{Perc} uses VGG16 [37] features. Metrics include PSNR, SSIM [43], LPIPS [56], and NIQE [34].

4.2 Comprehensive Quality Assessment and Performance Comparison

Having identified the limitations of various methods under UHD native inputs, we further adopt the inference strategies recommended in the official implementations of each baseline (e.g., overlapping patching or global downsampling) to evaluate their performance upper bounds under practically deployable conditions. Under these unified settings, we combine the quantitative metrics in the following tables with qualitative visual results to systematically analyze the image enhancement performance of different methods across multiple dimensions, including structural fidelity, perceptual quality, and detail restoration.

4.2.1 Quantitative Results Analysis. As shown in Tables 1 and 2, even when equipped with officially recommended compromise strategies, existing SOTA models still exhibit a severe "performance fragmentation" in UHD scenarios:

(1) **The latency trap of patching strategies:** Heavy networks like LLFormer [41] achieve a high PSNR of 37.31 dB at 4K, but they rely on overlapping patching at 8K due to strict VRAM constraints. This drastically increases inference time to approximately 170 seconds per image (0.006 FPS), making direct edge deployment prohibitive.

(2) **Inefficiency of patching strategies:** To achieve real-time processing, UHDM [51] and AGLLDiff [28] employ global spatial compression (e.g., resizing to 256×256) during 8K testing. This results in a irreversible loss of high-frequency details, causing their 8K PSNR to drop sharply to 14.30 dB and 11.80 dB, respectively, with all perceptual metrics degrading significantly.

(3) **Representational limits of lightweight networks:** FLOL [3], which performs native inference, manages to preserve a PSNR of 18.61 dB. However, its 421.2 ms latency still falls short of real-time processing standards.

In sharp contrast, our CPE model effectively decouples image quality from inference speed under end-to-end UHD input and output. In the 4K scenario, our method achieves a PSNR of 20.56 dB with a remarkably low latency of 8.68 ms (115.20 FPS). In the more extreme 8K scenario, it not only accomplishes native inference in 11.49 ms (87.00 FPS) but also delivers an excellent PSNR of 22.86

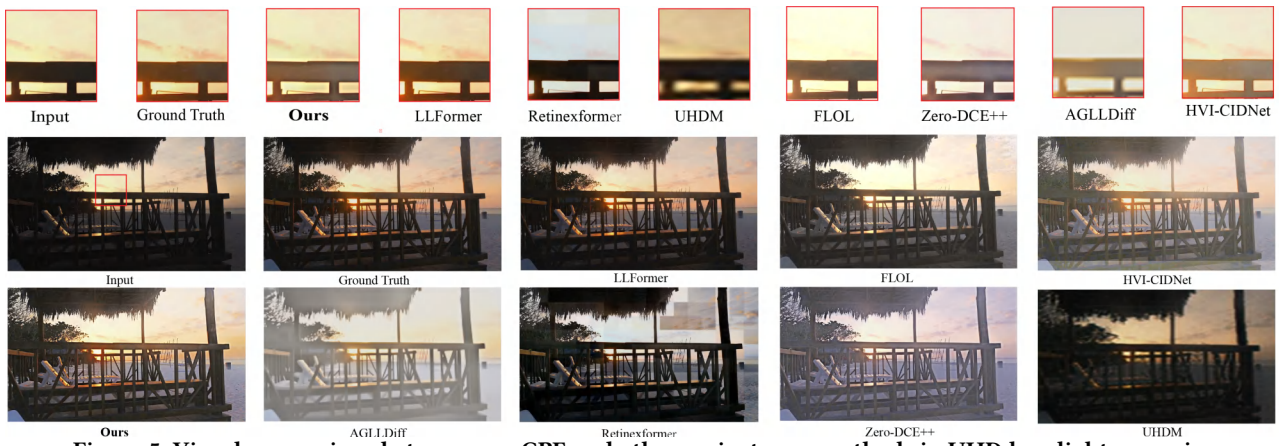


Figure 5: Visual comparison between our CPE and other mainstream methods in UHD low-light scenarios.

Table 1: Comparison of computational efficiency (Params, Inference Time, and FPS) in 4K and 8K UHD scenarios. The inference time is measured in milliseconds (ms). The best three results are highlighted in red, blue, and purple, respectively. Notice that heavy models like LLFormer [41] require over 160 seconds per 8K image, making them impractical for real-time applications.

Model	Params (M)↓	Strategy	4K (3840×2160)		8K (7680×4320)	
			Time (ms)↓	FPS↑	Time (ms)↓	FPS↑
LLFormer[41]	24.55	Native/Patching	35131.5	0.028	169990.5	0.006
HVI-CIDNet[49]	1.98	Native/Patching	456.79	2.19	1921.65	0.52
UHDM[51]	5.93	Native/Resize 256	451.9	2.21	24.13	41.43
Retinexformer[6]	1.61	Patching	426.9	2.34	3037.6	0.322
FLOL[3]	0.09	Native	109.0	9.17	421.2	2.37
Zero-DCE++[26]	0.01	Native / Scale	127.1	7.87	45.9	21.78
AGLLDiff[28]	552.98	Resize 256	8932.5	0.112	7555.4	0.132
Ours (CPE)	**1.14**	**Native E2E**	**8.68**	**115.20**	**11.49**	**87.00**

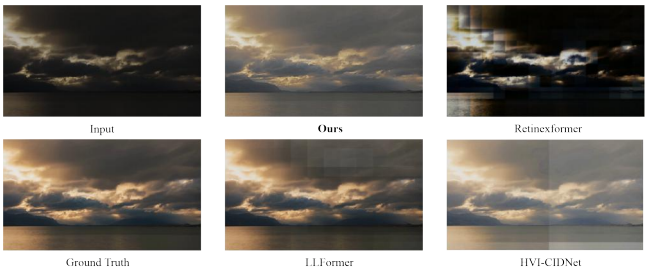


Figure 6: Evaluation of global illumination recovery in extremely dark wide-field scenes.

dB and an SSIM of 0.9453. Compared to both the "downsampling group" and the "native lightweight group," our method achieves superior results across efficiency and quality metrics, demonstrating promising potential for practical deployment.

4.2.2 Qualitative Visual Analysis. To further evaluate the differences in actual visual quality among the methods, we present the enhancement results in extreme low-light 4K and 8K scenarios in the following figures, alongside zoomed-in comparisons of highly challenging local textures. Overall, our proposed method exhibits distinct advantages in the following aspects:

Global Illumination Awareness and Spatial Fusion Consistency: As illustrated in Figure 6, heavy Transformer models (e.g., LLFormer [41]) are forced to employ overlapping patching inference due to VRAM constraints, which results in a severe loss of global illumination context. In regions with brightness gradients, such as the sky or walls, their enhanced results frequently exhibit abrupt "grid-like patch seams" and unnatural illumination discontinuities. In contrast, benefiting from the illumination decoupling design in the latent space, CPE achieves highly smooth and uniform global brightness elevation under full-resolution end-to-end inference, leaving the image completely free of any patching fragmentation.

High-Frequency Detail Fidelity at Native Resolution: When handling the massive spatial data of 8K images, models relying on global downsampling (e.g., UHDM [51] and AGLLDiff [28]) exhibit noticeable visual degradation. Due to severe spatial compression, these methods struggle to preserve the high-frequency textures of the original image. As a result, fine text edges and complex architectural structures suffer from smearing and blurring artifacts. In contrast, by applying the enhancement mapping directly to the native resolution, our CPE effectively maintains edge sharpness and structural clarity at full 8K resolution, avoiding the blurring issues caused by spatial downscaling.

Table 2: Quantitative comparison of low-light enhancement methods on 4K and 8K UHD datasets. 'Native' denotes original resolution input, 'Patching' denotes overlap sliding window strategies, and 'Resize' denotes global downsampling to avoid OOM. ↑/↓ indicate higher/lower is better. The best three results are highlighted in red, blue, and purple, respectively.

Model	Strategy	Metrics				
		PSNR↑	SSIM↑	VIF↑	LPIPS↓	NIQE↓
Panel A: Evaluation at 4K Resolution (3840×2160)						
LLFormer[41]	Native	37.31	0.9923	0.9036	0.0194	4.97
HVI-CIDNet[49]	Native	16.44	0.8776	0.7530	0.1702	6.78
UHDM[51]	Native	20.12	0.8968	0.5582	0.1428	5.39
Retinexformer[6]	Native	15.75	0.6851	0.6880	0.1949	4.35
FLOL[3]	Native	19.13	0.8291	0.7003	0.1608	4.56
Zero-DCE++[26]	Native	18.16	0.9087	0.6913	0.1622	5.08
AGLLDiff[28]	Resize 256	15.98	0.6933	0.1103	0.5453	12.63
Ours (CPE)	**Native E2E**	**20.56**	**0.9190**	**0.7872**	**0.1218**	**4.94**
Panel B: Evaluation at 8K Resolution (7680×4320)						
LLFormer[41]	Patching	34.70	0.9926	0.9038	0.0263	7.11
HVI-CIDNet[49]	Patching	17.04	0.8864	0.7530	0.1702	6.78
UHDM[51]	Resize 256	14.30	0.7068	0.0391	0.5817	15.83
Retinexformer[6]	Patching	16.85	0.7553	0.9238	0.2235	5.67
FLOL[3]	Native	18.61	0.9110	0.8026	0.2305	6.94
Zero-DCE++[26]	Scale ×12	19.37	0.9286	0.8657	0.2066	6.98
AGLLDiff[28]	Resize 256	11.80	0.6786	0.0628	0.5845	15.57
Ours (CPE)	**Native E2E**	**22.80**	**0.9453**	0.8542	**0.1345**	7.14

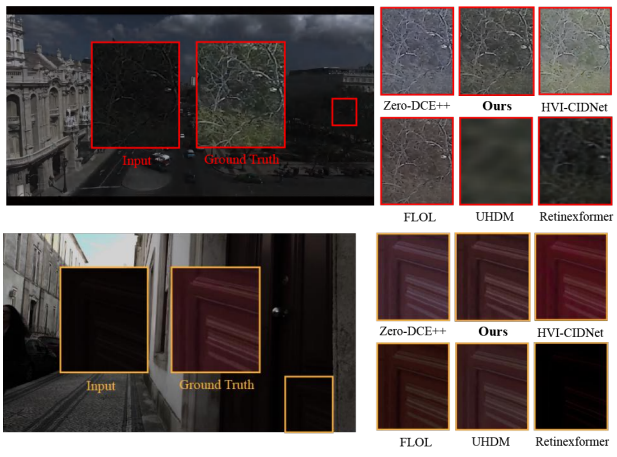


Figure 7: High-frequency detail reconstruction comparison at native 8K/4K resolution.

Color Constancy and Dark-Region Noise Suppression: For lightweight networks performing native inference (e.g., Zero-DCE++ [26]), their minimal parameter count creates representational bottlenecks when processing UHD images, making localized regions prone to over-enhancement and color shifts in dark areas. In contrast, driven by Retinex-based parameterized reconstruction and Clifford multivector fusion, our CPE effectively suppresses high-frequency noise in dark regions while substantially elevating brightness, faithfully restoring the authentic color ratios of the scene and delivering natural UHD visual results.

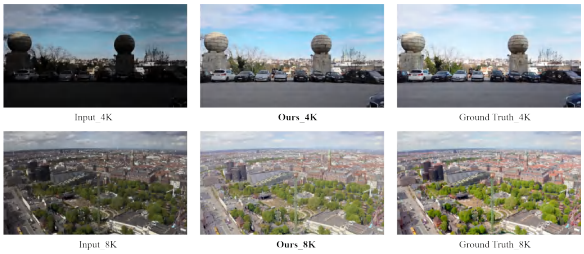


Figure 8: Demonstration of color constancy and global brightness enhancement by CPE at native 4K and 8K resolutions.

4.3 Ablation Study

To assess the contribution of each key component in CPE, we performed a series of incremental ablation studies under a unified benchmark. Starting from a minimalist U-Net as the baseline, we gradually integrated the proposed modules and evaluated their impact on both visual quality (PSNR, SSIM [43]) and computational efficiency (parameter count, memory footprint, inference latency, and FPS). Table 3 summarizes the detailed quantitative results.

Memory Footprint Reduction via Frequency Decoupling: Comparing Experiments 1 and 3, introducing the Frequency Decoupling (Freq-Sep) module into the baseline network increases the PSNR to 21.70 dB. Notably, the peak VRAM drops from 1143.16 MB to 768.29 MB. This confirms that decoupling low-frequency illumination from high-frequency textures effectively alleviates memory pressure during UHD image processing [11, 29].

Feature Conflicts and Retinex Mapping Correction: Naively stacking modules can lead to feature conflicts and semantic misalignments [48]. As shown in Experiment 5, simultaneously introducing the multi-scale pyramid and frequency decoupling without feature alignment causes performance to degrade, with the PSNR falling to a baseline low of 20.89 dB. However, adding the Retinex Illumination Mapping (Illu-Map) in Experiment 7 recovers and boosts the PSNR to 21.77 dB. This suggests that physics-constrained parameter mapping effectively corrects multi-scale feature misalignment, guiding the network toward accurate enhancement.

Performance Leap via Clifford Geometric Fusion: Achieving robust, high-quality feature fusion is a core objective of this work. Comparing Experiment 7 with the final full model (Experiment 8), integrating the Clifford Geometric Fusion module yields a substantial performance leap. The PSNR jumps by nearly 1 dB, from 21.77 dB to 22.76 dB, and the SSIM improves to 0.9437. Crucially, this visual quality gain comes at a negligible cost: zero additional parameters (remaining constant at 1.144 M) and a marginal latency increase of only 0.19 ms. These results confirm that geometric fusion within the multivector space [4] effectively overcomes the feature aggregation bottleneck in UHD images.

4.4 Downstream Task Evaluation

To validate CPE's practical utility for downstream high-level vision tasks [14, 33], we conducted object detection on the 4K/8K test subsets of UHD-LOL [41] using a pre-trained YOLOv10x [39] (without low-light fine-tuning). During testing, the enhanced outputs from various baseline models (e.g., LLFormer [41], UHDM [51], AGLLDiff [28]) and our CPE were directly fed into the detector to count total detections and calculate mean confidence for core categories.

Table 3: Quantitative results of the ablation study. We progressively integrate the Multi-scale Pyramid (Pyramid), Frequency Decoupling (Freq-Sep), Retinex Illumination Mapping (Illu-Map), and Clifford Geometric Fusion (Clifford) into the baseline network. Evaluation is performed on the 8K UHD dataset.

Exp.	Proposed Components					PSNR↑	SSIM↑	Params (M)↓	VRAM (MB)↓	Latency (ms)↓	FPS↑
	Baseline	Pyramid	Freq-Sep	Illu-Map	Clifford						
1	✓					21.56	0.9212	0.434	1143.16	11.36	88.00
2	✓	✓				21.58	0.9313	0.572	1145.34	12.01	83.27
3	✓		✓			21.70	0.9297	0.868	768.29	12.26	81.53
4	✓		✓		✓	21.61	0.9274	0.868	771.60	12.01	83.24
5	✓	✓	✓			20.89	0.9326	1.143	1156.35	13.47	74.25
6	✓		✓	✓	✓	21.52	0.9312	0.873	1159.68	12.92	77.37
7	✓	✓	✓	✓		21.77	0.9344	1.144	1164.06	13.57	73.68
8	✓	✓	✓	✓	✓	**22.76**	**0.9437**	1.144	1168.44	13.76	72.66

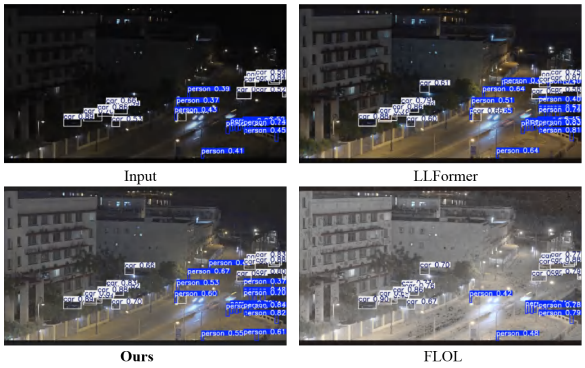


Figure 9: Downstream nighttime object detection results using YOLOv10x. Bounding boxes indicate detected pedestrians, vehicles, and their confidence scores.

Supported by visual results (Figure 9) and quantitative metrics, CPE demonstrates robust 8K small-object detection, identifying 576 targets—far exceeding UHDM (0) and AGLLDiff (35). Moreover, CPE yields an 8.7% gain in 4K detections over raw inputs (2264 vs. 2082), with a mean confidence of 0.7841.

Regarding computational efficiency, while LLFormer achieves the highest detection count (2296), its 8K inference latency of nearly 170 seconds per frame makes it impractical for real-world deployment. In contrast, CPE achieves a 14,000× speedup (11.49 ms) while maintaining 98.6% of LLFormer's detection capability. This demonstrates that CPE not only provides sharp and physically consistent inputs for high-level vision tasks but also offers a significant advantage for the real-time deployment of UHD vision systems [45].

4.5 Mobile Edge Deployment

To validate CPE's real-world applicability on resource-constrained edge devices [21, 47], we deployed an FP16 mixed-precision prototype [12] on commercial smartphones (Huawei Mate 60 Pro and iPhone 16 Pro) equipped with an embedded NPU. Benefiting from our "low-resolution estimation, high-resolution mapping" strategy, CPE fundamentally bypasses the memory bottlenecks of massive UHD convolutions. For instance, when tested on a Huawei Mate 60 Pro, the core network pure NPU execution takes only ~279 ms, while the entire end-to-end pipeline (including I/O and rendering) runs in just ~312 ms. This demonstrates CPE's capability to deliver real-time high-quality low-light enhancement directly on mobile platforms, a significant practical advantage over heavy models requiring cloud GPU clusters [6, 41].

5 Discussion and Limitations

By embedding geometric constraints via Clifford algebra, CPE preserves spatial directionality and reduces texture misalignment without significant regularization overhead. Additionally, our multi-scale Retinex mapping efficiently processes 4K/8K inputs, circumventing typical out-of-memory issues. However, we still have room for improvement. For instance, relying on Retinex priors may occasionally cause slight color halos around intense point lights (e.g., vehicle high beams). And if we directly extend this single-frame architecture to UHD video streams, minor flickering might appear in practice [55]. Finally, regarding our evaluation, although the UHD enhancement community has seen inspiring and rapid progress in early 2026[59], to ensure rigorous and reproducible hardware stress tests, we restrict our evaluation to fully open-source baselines. We look forward to broader cross-architecture comparisons as concurrent works release their code.

6 Conclusion

This paper presents CPE, a high-speed low-light enhancement architecture for UHD images based on Clifford geometric feature fusion. By employing a strategy of "latent space decoupling and geometric fusion at low resolution, followed by native parameter mapping at high resolution," we circumvent edge artifacts and VRAM overflow risks at the structural level. Experiments have demonstrated that CPE achieves a remarkable 11.49 ms inference speed and peak memory consumption (1.22 GB) at 8K resolution, while faithfully restoring UHD textures and colors in extreme darkness. Our method provides both a theoretical reference and a feasible technical path for real-time deployment of UHD vision on resource-constrained edge terminals. Future work will explore generative priors to hallucinate these extreme blind spots, alongside introducing 3D spatio-temporal geometric manifolds to optimize temporal consistency in dynamic video streams.

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